Supporting Information

Coupling iron sludge addition and intermittent aeration for achieving simultaneous methanogenesis, Feammox and denitrification in a single reactor treating fish sludge

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7 pages (page S1-page S7), one table (Table S1) and five figures (Figure S1-Figure S5) are contained in the Supporting Information.

Table S1. Main characteristics of the fish sludge, inoculum, and mixture.

Parameter	Fish sludge	Inoculum	Mixture [a]
Total solids (TS, g/L)	86.2 ± 0.2	61.0 ± 0.1	50.30 ± 0.2
Volatile solids (VS, g/L)	28.4 ± 0.1	20.6 ± 0.1	29.90 ± 0.1
Total nitrogen (TN, mg/L)	1578.8 ± 34.5	313.5 ± 12.3	1197.7 ± 30.7
Ammonia nitrogen (NH_4^+-N, mg/L)	288.8 ± 14.7	58.9 ± 3.6	143.1 ± 9.4
Nitrate nitrogen (NO_3^--N, mg/L)	49.4 ± 2.8	0.21 ± 0.0	18.64 ± 1.2
Nitrite nitrogen (NO_2^--N, mg/L)	—— [b]	——	——
Total chemical oxygen demand (TCOD, mg/L)	26848.8 ± 764.2	18840.0 ± 643.8	22166.5 ± 830.7
Soluble chemical oxygen demand (sCOD, mg/L)	3017.1 ± 22.1	151.2 ± 9.7	1718.3 ± 28.8
Soluble carbohydrates (PS, mg/L)	478.4 ± 19.8	55.6 ± 1.8	223.7 ± 8.8
Soluble proteins (PN, mg/L)	432.3 ± 21.2	72.5 ± 1.9	212.7 ± 11.9
C/N ratio	16.5	13.2	15.7
pH	7.5 ± 0.1	7.1 ± 0.1	7.0 ± 0.1

* All results are presented as average values ± standard deviation based on three tests.

[a] Fish sludge and inoculum were mixed at a VS ratio of 3:1.

[b] Under detection limit.

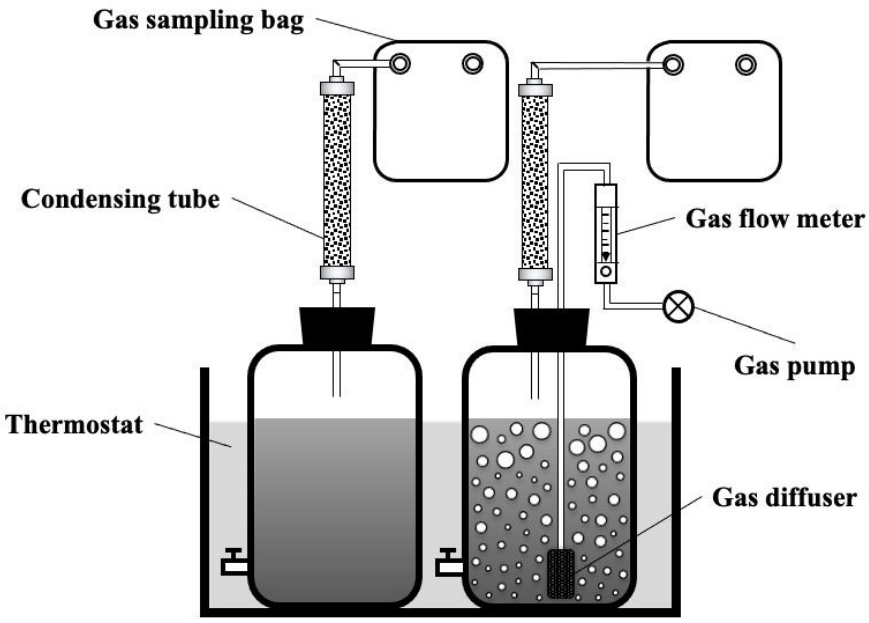


Figure S1. Schematic of the reactors used in this study.

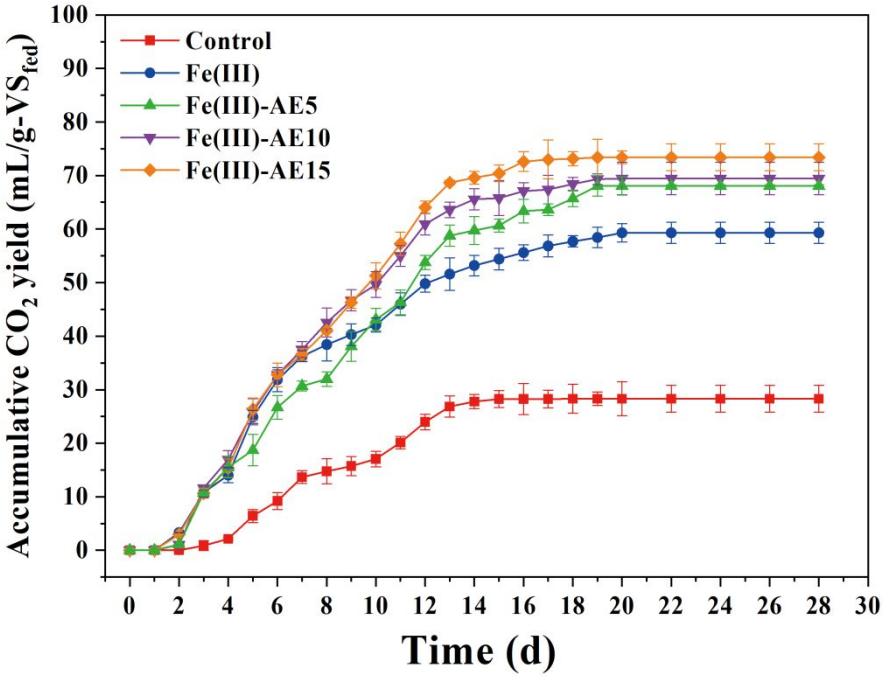


Figure S2. The accumulative CO_2 yield in different reactors.

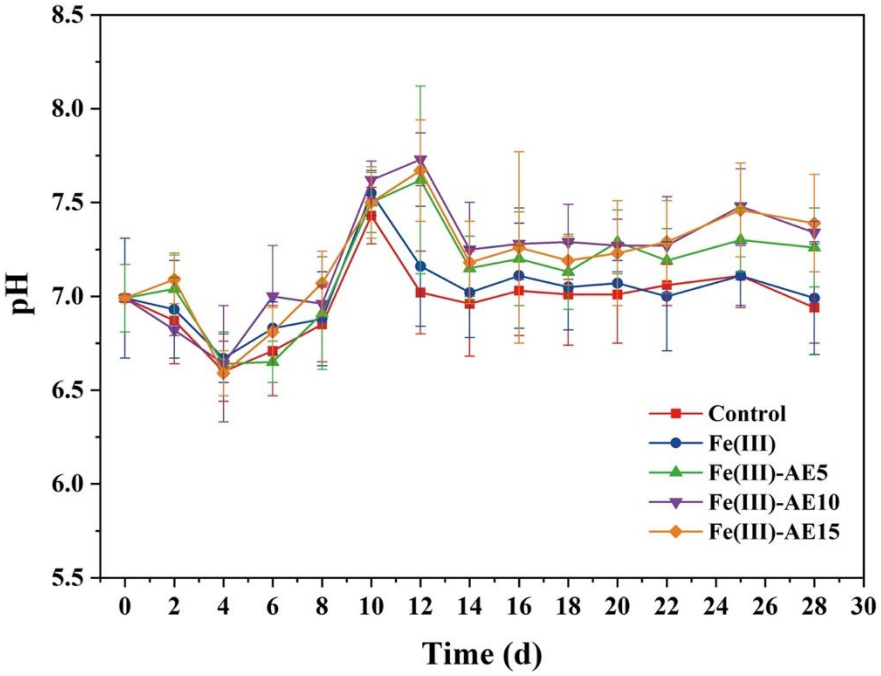


Figure S3. The variations of pH in different groups during AD.

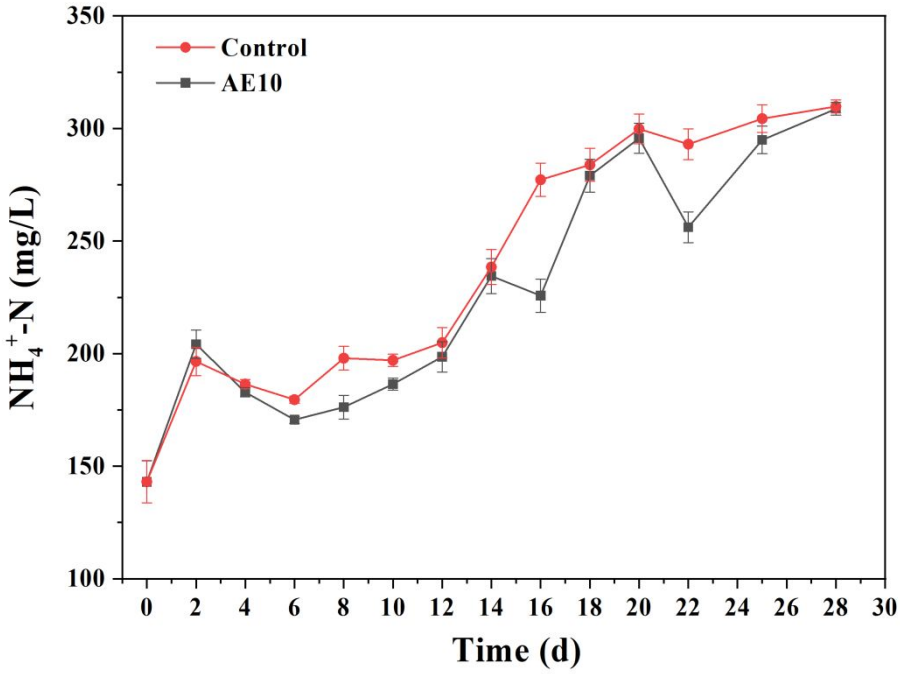


Figure S4. Changes of NH_4^+-N in the control and AE10 groups.

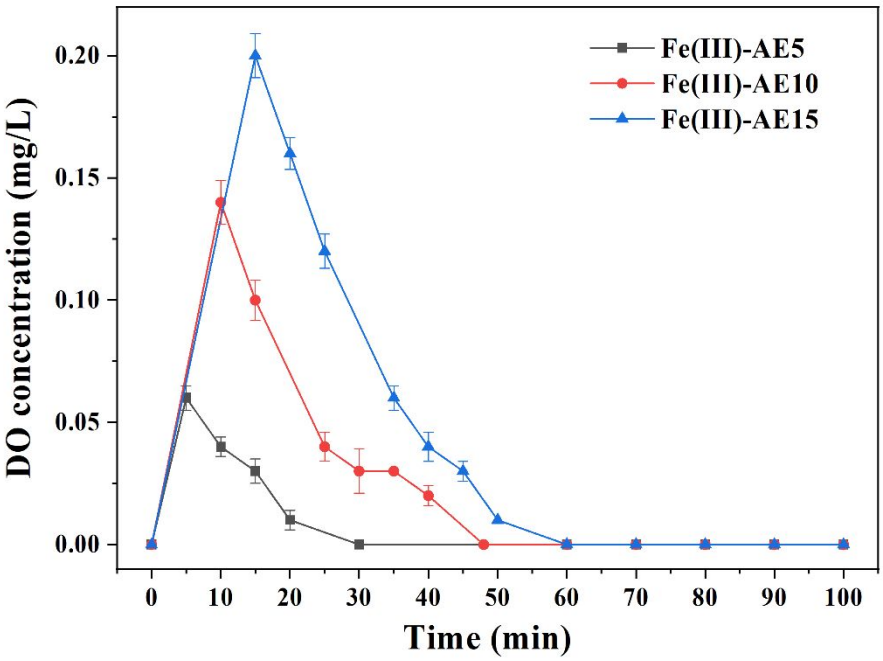


Figure S5. The changes of DO during one aeration cycle on day 4.